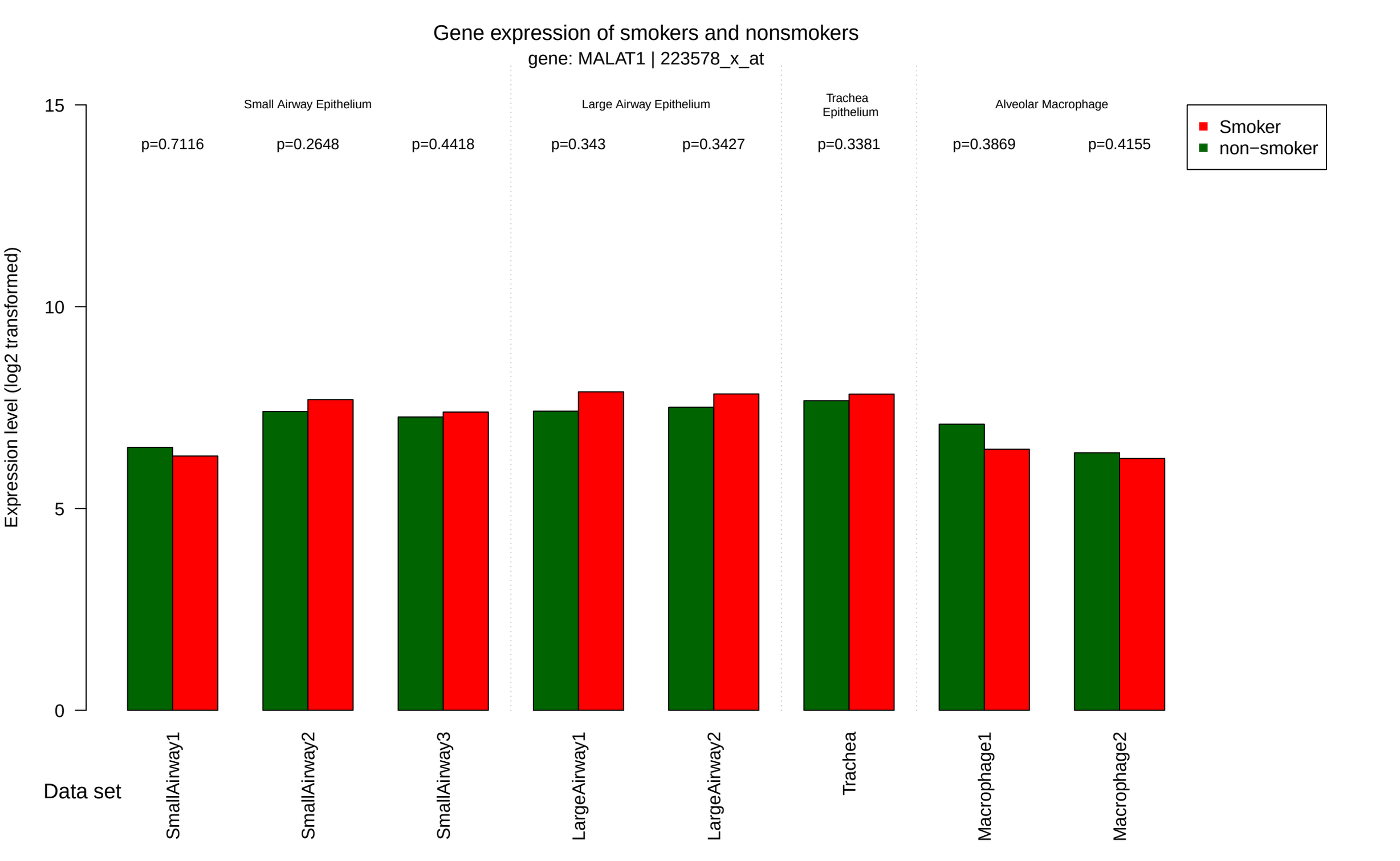
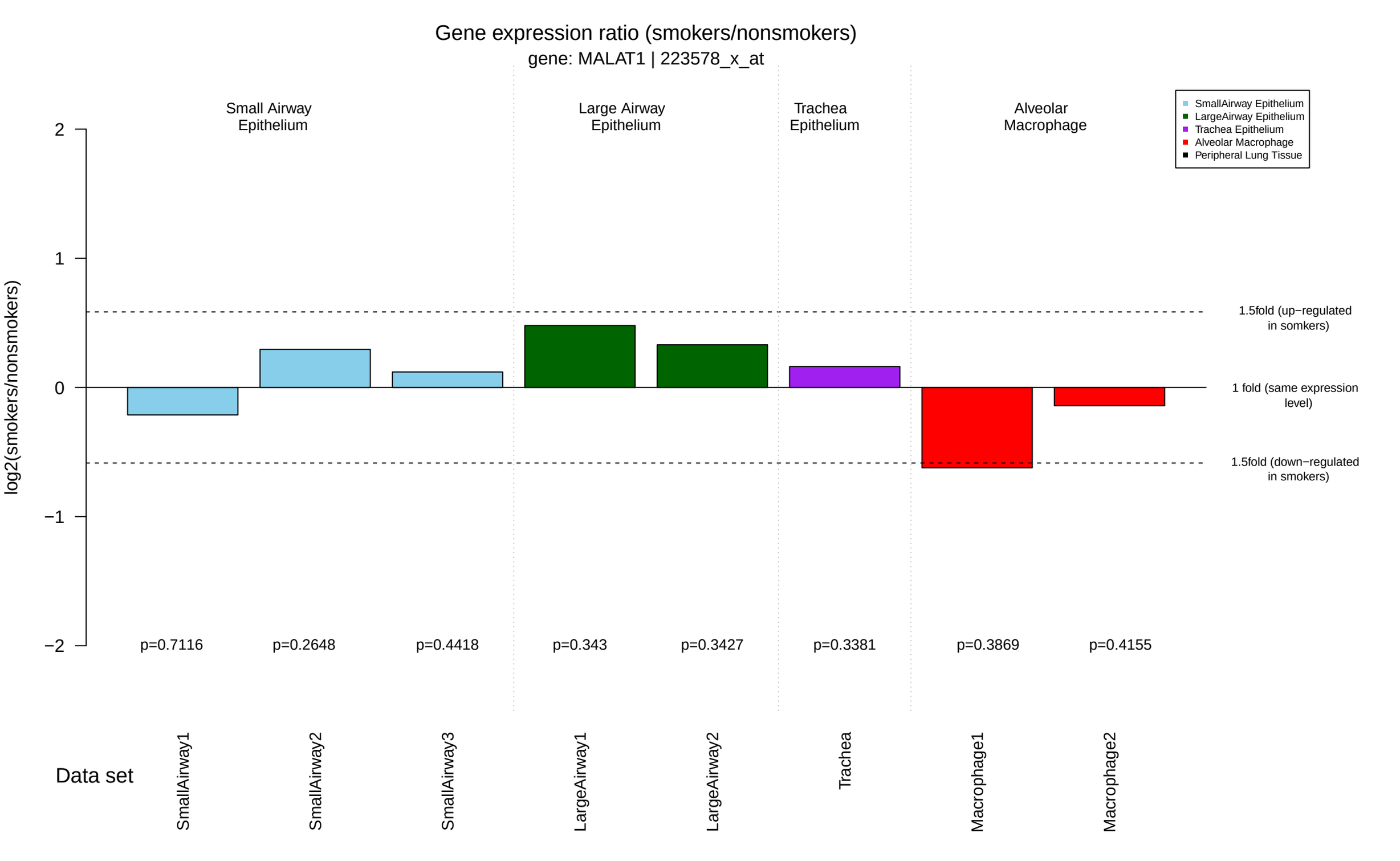
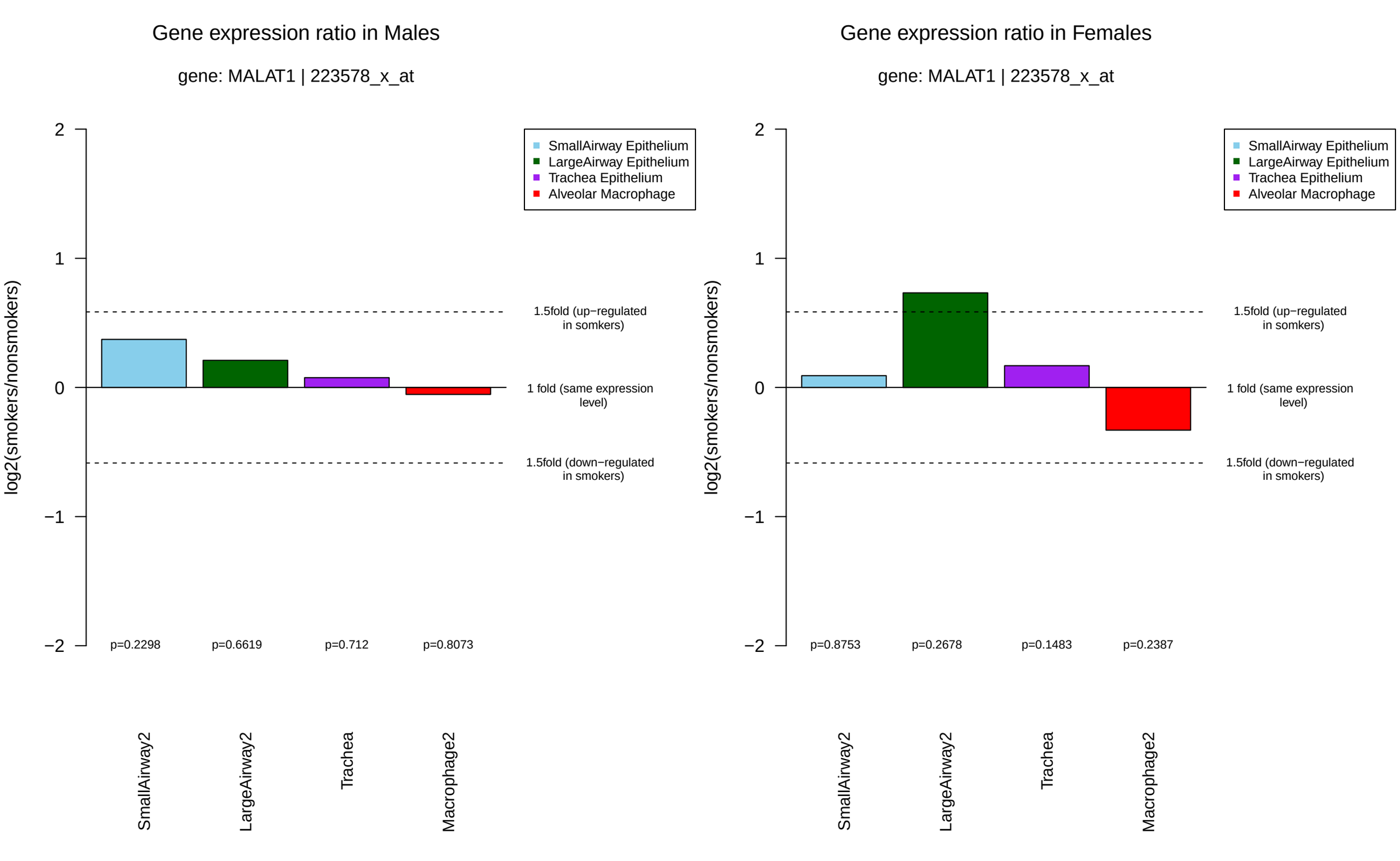
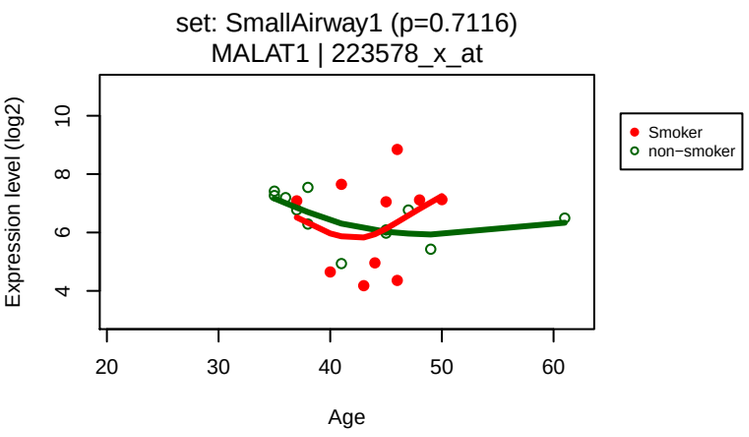
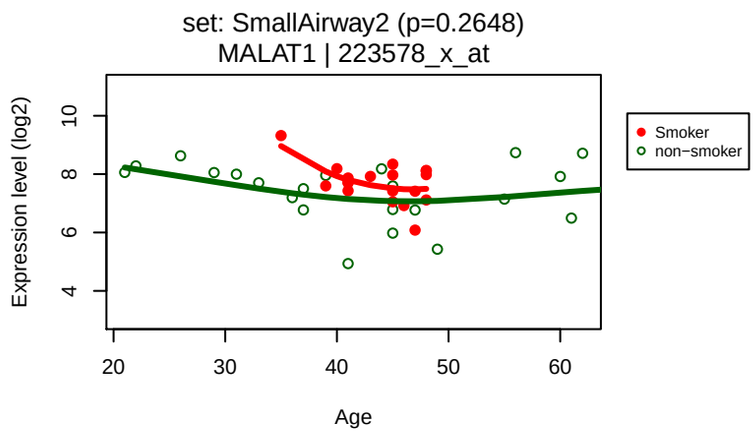
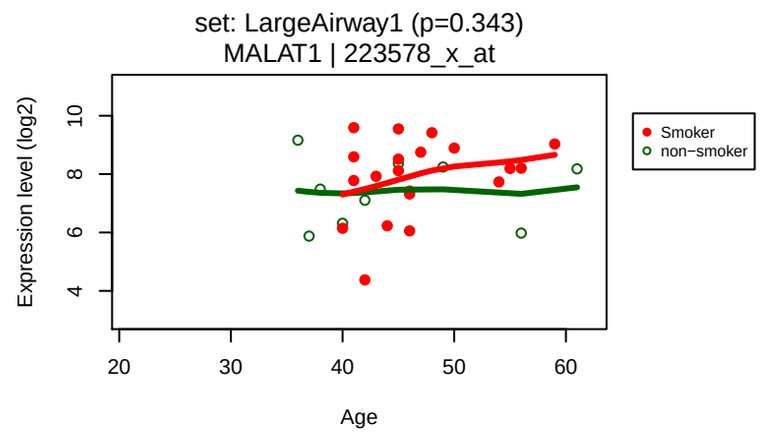

Gene expression of smokers and nonsmokers
gene: MALAT1 | 223578_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7116
p=0.2648
p=0.4418
p=0.343
p=0.3427
p=0.3381
p=0.3869
p=0.4155
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

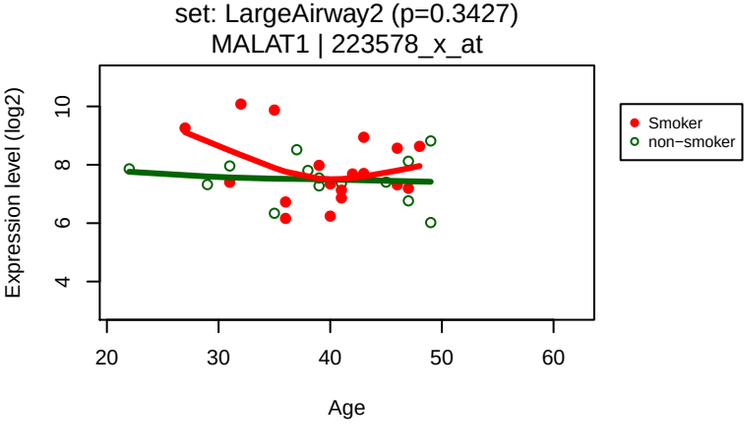

Gene expression ratio (smokers/nonsmokers)
gene: MALAT1 | 223578_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7116
p=0.2648
p=0.4418
p=0.343
p=0.3427
p=0.3381
p=0.3869
p=0.4155
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

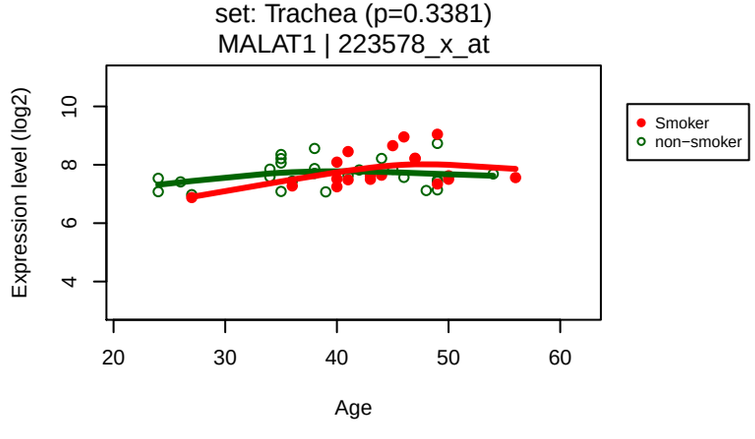

Gene expression ratio in Males
gene: MALAT1 | 223578_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2298
p=0.6619
p=0.712
p=0.8073
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MALAT1 | 223578_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8753
p=0.2678
p=0.1483
p=0.2387
SmallAirway2
LargeAirway2
Trachea
Macrophage2

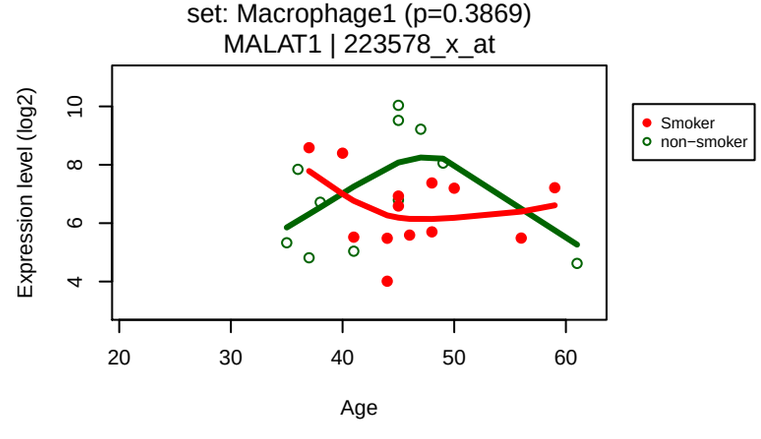

set: SmallAirway1 (p=0.7116)
MALAT1 | 223578_x_at
Expression level (log2)
Age
Smoker
non−smoker

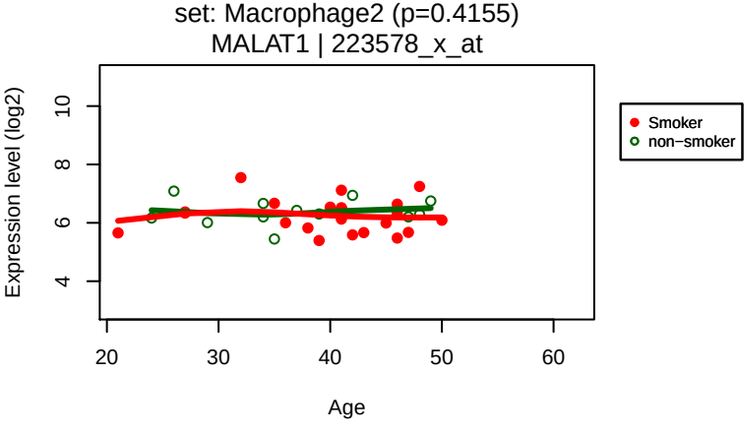

set: SmallAirway2 (p=0.2648)
MALAT1 | 223578_x_at
Expression level (log2)
Age
Smoker
non−smoker

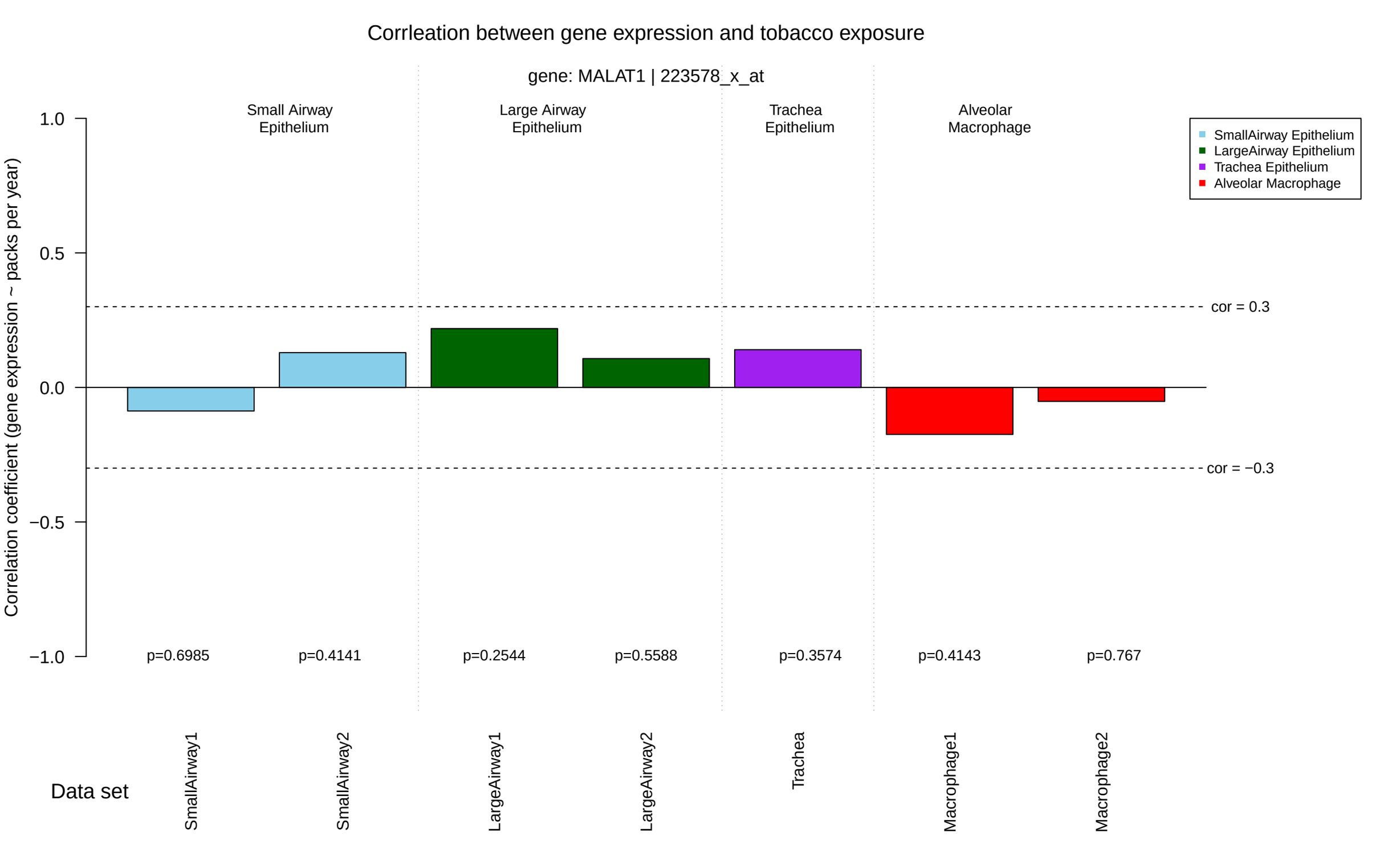

set: LargeAirway1 (p=0.343)
MALAT1 | 223578_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3427)
MALAT1 | 223578_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3381)
MALAT1 | 223578_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3869)
MALAT1 | 223578_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4155)
MALAT1 | 223578_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MALAT1 | 223578_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6985
p=0.4141
p=0.2544
p=0.5588
p=0.3574
p=0.4143
p=0.767
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2